

A TOUR OF C++

MIHIR RAO

Contents

1	The Basics	2
1.1	Programs and compilation	2
1.2	Functions	2
1.3	Types, objects, and arithmetic	3
1.4	Initialization and names	3
1.5	Constants and compile-time evaluation	4
1.6	Arrays, pointers, and references	4
1.7	Tests and loops	5
1.8	The memory model: copying and binding	5
1.9	Practical habits	6

CHAPTER 1: THE BASICS

C++ programs combine a small set of built-in operations with library components and types supplied by programmers. The core language gives direct access to memory and computation; functions and types give those operations useful names.

1.1 Programs and compilation

A compiler translates each SOURCE FILE into an OBJECT FILE. A LINKER combines object files and required library code into an executable for a particular platform. Source code can be portable across platforms even though a compiled executable usually is not.

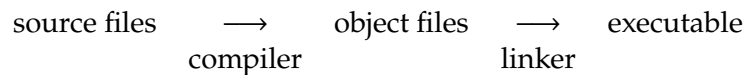


Figure 1: The two stages that turn C++ source into a program.

The STANDARD LIBRARY supplies facilities such as streams and containers. Its declarations must be made available before use. The familiar header form works widely; implementations that support the standard-library module can instead use `import std;`.

```
1 #include <iostream>
2 int main()
3 {
4     std::cout << "Hello, C++!\n";
5 }
```

Execution begins at the program's single global `main` function. `std::cout` is the standard output stream, and `<<` sends the string to it. The `std::` qualification identifies the library namespace. A zero return from `main`, including an implicit return at its end, indicates success. The compiler knows the TYPE of every entity at its point of use, so it can check which operations are valid.

1.2 Functions

A FUNCTION DECLARATION states a name, a return type, and parameter types. A FUNCTION DEFINITION also supplies the body. A call needs a visible declaration, and its arguments are initialized from the supplied expressions. `void` means that no value is returned.

```
1 double square(double x);           // declaration
2 double square(double x) { return x*x; } // definition
3 double area(double r) { return 3.14159*square(r); }
```

Breaking a computation into short, named functions makes its steps and dependencies visible. **OVERLOADING** gives several functions the same name when their parameter types differ and their purpose is the same. The compiler chooses the best match for a call; a call with equally good matches is ambiguous and rejected.

1.3 Types, objects, and arithmetic

A type determines a set of values, permitted operations, and the representation of an **OBJECT**, a region of storage holding a typed value. A **VARIABLE** is a named object. Fundamental types include `bool`, `char`, `int`, and `double`. Their exact sizes depend on the implementation: `sizeof(char)` is 1, while `sizeof(int)` is often 4. An unsigned integer is primarily useful for bit operations, not as a general way to rule out negative numbers.

Integer literals are decimal unless marked otherwise: `0b1010` is binary, `0xA` is hexadecimal, and a leading `0` denotes octal. A decimal point or exponent makes a floating-point literal, as in `3.14` or `314e-2`. A single quote can separate digits without changing the value, as in `1'000'000`.

Arithmetic uses `+`, `-`, `*`, `/`, and, for integers, remainder `%`. Comparisons include `==` and `!=`; `=` assigns a value. The logical operators `&&`, `||`, and `!` differ from the bitwise operators `&`, `|`, `^`, and `~`. Mixed arithmetic can convert operands, for example an `int` to a `double`. Converting the result back to an integer can discard its fractional part. Keep expressions simple when conversions or evaluation order could obscure their meaning.

1.4 Initialization and names

INITIALIZATION creates an object with its first value; **ASSIGNMENT** replaces the value of an existing object. A brace initializer rejects **NARROWING CONVERSIONS** that would lose information:

```
1 double rate {2.5};
2 int a = 2.5;      // allowed, but loses the fractional part
3 int b {2.5};      // error: narrowing
4 auto count = 12;  // deduced as int
```

Prefer braces for a declared type and initialize each variable when it is introduced. `auto` asks the compiler to deduce a type from the initializer; it is especially helpful when the type is obvious or

cumbersome to spell. State the type when its range, precision, or meaning needs emphasis.

A name has a `SCOPE`, the region where it can be used. A local name lasts from its declaration to the end of its enclosing block. A class member belongs to class scope, and a namespace member belongs to namespace scope. The `LIFETIME` of an ordinary local object ends when its block exits; a namespace object lasts until program termination. A dynamically allocated object instead lasts until it is explicitly destroyed. Keep names in the smallest useful scope.

1.5 Constants and compile-time evaluation

`const` prevents modification through a name, even when its initial value is computed at run time. `constexpr` requires an initial value that the compiler can evaluate. A `constexpr` function may also run with ordinary run-time arguments, whereas a `constexpr` function must be evaluated at compile time.

```
1 constexpr int cube(int x) { return x*x*x; }
2 constexpr int twice(int x) { return 2*x; }

3 int input = 3;
4 const int observed = cube(input);    // runtime value, then immutable
5 constexpr int fixed = cube(3);      // compile-time value
6 int scaled = cube(input);           // valid runtime call
7 int error = twice(input);           // error: input is not constant
```

Constant expressions are required in places such as built-in array bounds and case labels. Immutability also makes ordinary code easier to reason about.

1.6 Arrays, pointers, and references

An `ARRAY` stores a fixed number of elements contiguously and uses zero-based indices. A `POINTER` holds an address; `&` obtains an object's address and `*` dereferences a pointer. In the following example, `p` points to the fourth element.

```
1 char letters[6] {'a', 'b', 'c', 'd', 'e', 'f'};
2 char* p = &letters[3];
3 char fourth = *p;           // 'd'
```

A `RANGE-FOR LOOP` traverses all elements without an explicit index. Its loop variable can copy each element or refer to it:

index	0	1	2	3	4	5	
letters	a	b	c	d	e	f	$p = \&\text{letters}[3]$

Figure 2: A pointer identifies one element within a contiguous array.

```

1 int values[] {2, 4, 6};
2 for (auto x : values) { std::cout << x << '\n'; } // copies
3 for (auto& x : values) { ++x; } // modifies elements

```

A REFERENCE is bound to an object at initialization and cannot be rebound. Passing a reference lets a function modify the caller's object without copying it; passing a const reference avoids a copy while promising not to modify it. A reference must refer to a valid object.

Use `nullptr` when a pointer intentionally has no target. Check it before dereferencing. A pointer to a zero-terminated character array may be advanced one element at a time until it reaches the terminating zero; this requires that the array really has such a terminator. `nullptr` is a pointer value, unlike the integer `0` used in older code.

1.7 Tests and loops

An `if` chooses a branch, a `switch` selects among distinct constant case labels, and `while` and `for` repeat work. `break` exits a loop or switch. A test of a number asks whether it is nonzero; a test of a pointer asks whether it is non-null. An `if` can declare a value whose scope stays within the statement:

```

1 std::vector<int> items {2, 4, 6};
2 if (auto n = items.size(); n != 0) {
3     std::cout << n << " elements\n";
4 }

```

Choose a range-for loop for straightforward traversal and keep temporary names near their first use.

1.8 The memory model: copying and binding

Built-in operations usually map closely to hardware instructions, and pointers expose machine addresses through typed operations. Assignment copies a value into an existing object. Copying one integer does not make two names refer to the same integer; assigning one pointer to another copies an address, so both pointers may then point to the same object.

```
1 int x {2};
2 int y {3};
3 int* p {&x};
4 int* q {&y};
5 p = q;           // both pointers now point to y; x is unchanged

6 int& r {x};
7 int& s {y};
8 r = s;           // copies y's value into x; r still refers to x
```

The distinction between binding and assignment matters for references. `int& r{x}` initializes a reference by binding it to `x`; later `r = s` assigns through the reference. An uninitialized reference is not allowed. Function argument passing and return values also use initialization semantics.

1.9 Practical habits

Build programs from appropriately named functions and library facilities. Initialize objects promptly, avoid narrowing, prefer immutable values when possible, and use `nullptr` for absent pointers. Keep pointer use simple, choose range-for loops for full traversal, and write comments that explain intent rather than restating code.

Source. Bjarne Stroustrup, *A Tour of C++*, third edition (Pearson, 2023), Chapter 1, pp. 1–20.